HVAC Diagnostic & Repair Report

Service Provider: Apex Climate Solutions LLC

License Number: #HVAC-99281-B

Technician Name: Marcus Vance

Date of Inspection: October 17, 2026

Work Order / Claim Ref: #HW-88301-A

1. Equipment Information

System Type: Split System Air Conditioner (Condenser)	**Manufacturer:** Carrier
Model Number: 24ABC636A003	**Serial Number:** 1416E22894
Age of Unit: 10 Years	**Location:** Exterior Rear Yard

2. Diagnostic Findings & Cause of Failure

Upon arrival, the homeowner reported that the system was running but blowing room-temperature air. An electrical and mechanical diagnostic on the exterior condenser unit revealed the following:

• **Compressor Assessment:** The compressor is drawing locked rotor amps (LRA) at startup, cycling heavily on internal thermal overload, and failing to compress refrigerant.

• **Electrical Check:** The dual run capacitor and contactor were tested; both are functioning within normal microfarad and voltage parameters.

• **Refrigerant Pressures:** Static pressures are stable, ruling out an active refrigerant leak as the cause of the failure.

• **Cause of Failure:** Internal mechanical failure of the compressor due to normal operational wear and tear over a 10-year lifespan. No signs of homeowner neglect, lack of maintenance, or pre-existing physical damage were observed.

3. Itemized Repair Estimate

Item / Description	Qty	Rate	Total
OEM Replacement Compressor Carrier 3-Ton R-410A	1	$950.00	$950.00
Filter Drier Replacement Required for system open	1	$75.00	$75.00
R-410A Refrigerant Recovery & Recharge Up to 6 lbs	1	$325.00	$325.00

Labor System evacuation, brazing, vacuum pull, and testing	4 Hrs	$112.50	$450.00
		GRAND TOTAL	**$1,800.00**

4. Technician Notes & Recommendations

The internal components of the compressor are seized. Repair requires recovering the existing refrigerant, cutting out the failed compressor, brazing in the new OEM unit, installing a new liquid line filter drier, pulling a vacuum down to 500 microns, and recharging the system to factory specifications. The system coil and overall ductwork remain in good condition. A repair is highly recommended over a full system replacement at this time.